

Mohammed Al-Rubaiai

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PROFESSIONAL PROFILE

High-impact Staff Electrical Engineer specializing in high-density mechatronic system architecture, advanced sensor integration, and hardware-software co-design for complex robotic platforms and custom data-collection wearable devices. Driven by first-principles engineering, possesses a proven track record of owning the full hardware lifecycle—spanning component selection, dense rigid-flex PCB layout, bring-up, and vendor-managed mass production. Expert at bridging electrical, mechanical, and firmware boundaries to design ultra-compact sensing solutions and wearable data-collection systems that capture high-fidelity datasets to train AI models and solve critical sim-to-real bottlenecks.

PROFESSIONAL EXPERIENCE

Apptronik

Staff Electrical Engineer – Sensor Integration

Austin, TX

Jan 2026 – Present

- **Architecting the humanoid hand sensing subsystem**, defining end-to-end electrical architectures, dense PCB layouts, and robust interconnect schemes utilizing rigid-flex arrays, stretchable conductors, and micro-coaxial cabling.
- **Implementing ultra-compact, high-bandwidth data pipelines** within severe mechanical constraints, leveraging touch controller ASICs and implementing SPI, I2C, I3C, and Single Twisted-Pair Ethernet (SPE) protocols.
- **Spearheading the co-design of a novel wearable data-collection device** in partnership with Mechanical, Firmware, Material, and AI teams, establishing hardware specifications to capture high-fidelity manipulation datasets and bridge the sim-to-real gap.
- **Support the design of high-density BLDC motor driver boards** for the forearm actuator package, managing critical sensor selection and layout strategies to guarantee signal integrity and absolute position feedback.

Clone Robotics

Robotics Sensing Consultant

Palo Alto, CA (Remote)

Sep 2025 – Dec 2025

- **Conducted a comprehensive state-of-the-art trade study** evaluating commercial and academic tactile modalities, establishing an architectural pros-and-cons framework to guide platform-level sensor selection.
- **Designed and prototyped an ultra-thin 0.2 mm flexible capacitive sensor** integrated with custom readout electronics for high-resolution joint angle measurement.
- **Developed an instrumentation solution** to accurately measure the real-time extension and compression metrics of biomimetic pneumatic actuators.

Tesla

Senior Robotics Hardware Engineer

Palo Alto, CA

Feb 2024 – Aug 2025

- **Developed Tesla's first 3D-shaped multi-modal (capacitive and resistive) tactile sensor**, owning the full lifecycle from raw material selection and in-house functional prototyping to establishing mass-production workflows with external vendors.
- **Co-designed, validated, and characterized the second-generation flat tactile sensors** integrated across the Optimus humanoid hand fingertips, directly supporting associated dense PCB layouts and ruggedized interconnect schemes.
- **Headed the development and validation of the foot load cell systems**, owning the electrical board design, sensor integration, and structural validation protocols to ensure accurate ground-reaction force feedback.
- **Executed rapid bench-top debugging and root-cause failure analysis** on active Optimus fleets, identifying design flaws and rolling out targeted hardware resolutions to boost reliability.

ByteDance

Haptic Soft Actuator Scientist

Greater Seattle Area

Aug 2022 – Dec 2023

- **Led the end-to-end prototyping and characterization** of multimodal soft transducers for VR/AR input devices, resulting in **2 granted US patents** for adaptive triggers and haptic rotary inputs.

- **Validated novel foam-based magnetic tactile sensors**, culminating in a peer-reviewed publication demonstrating sub-millimeter spatial sensing integrated into multiple spatial computing form factors.
- **Coordinated cross-functional R&D initiatives** spanning materials science, mechanical design, electronics, and UX to transition soft actuator concepts into wearable consumer hardware prototypes.

Toyota North America
Research Scientist

Ann Arbor, MI
Nov 2021 – Aug 2022

- **Led R&D initiatives** focused on soft robotics, smart materials, and mechatronic systems for wearable applications, utilizing data-driven modeling and advanced control strategies.
- **Accelerated technology transfer** by introducing innovative polymer technologies to core mechatronics platforms, contributing high-value technical insights to future product pipelines.

Michigan State University
Research Assistant

Lansing, MI Area
Aug 2016 – Dec 2021

- **Conducted strategic research** in additively manufactured soft actuators, stiffness tuning, and lightweight 3D-printed strain and pressure sensors.
- **Supported sponsored industrial projects for Toyota** through comprehensive modeling and characterization of PVC gel actuators, resulting in **1 granted US patent**.

Schlumberger
Wireline Field Engineer

Basra, Iraq
Apr 2012 – May 2013

- Managed field crews and high-value wireline assets to execute complex data acquisition jobs under strict client specifications and strict safety compliance guidelines.

Taehwa
Control Engineer

Erbil, Iraq
Sep 2009 – Oct 2011

- Programmed and debugged PLCs, diagnosed complex electromechanical issues, and validated industrial computer systems for automated industrial equipment.

EDUCATION & CREDENTIALS

Ph.D. in Electrical & Computer Engineering
M.S. in Electrical & Computer Engineering
B.S. in Mechatronics Engineering

Michigan State University, 2021
Michigan State University, 2015
University of Baghdad, 2009

PATENTS & PUBLICATIONS SUMMARY

- **Patents:** Granted 3 US Patents as primary/co-inventor covering flexible sensor arrays, adaptive triggers for soft actuators, and haptic rotary inputs (*US12055449*, *US12056282*, *US12189858*).
- **Publications:** Author/Co-author of **15+ peer-reviewed papers** in high-impact journals and conferences including *IEEE/ASME Transactions on Mechatronics*, *Soft Robotics*, and *Smart Materials and Structures*.
- **Editorial Contributions:** Technical reviewer for top-tier robotics and smart materials IEEE/ASME journals.

HONORS, AWARDS & RECOGNITIONS

- **EB-2 National Interest Waiver (NIW):** Approved Permanent Residency based on significant research contributions to soft robotics and sensor technologies.
- **The Best Student Paper Gold Award:** 2021 Modeling, Estimation and Control Conference (MECC).
- **Fellowships:** Awarded the MSU Dissertation Completion Fellowship (2021) and Graduate School Fellowships (2015–2017).